

AI/ML Adaptive Noise Cancellation for Defence Communications

• **Problem Statement ID – SIH26052**

• **Problem Statement Title-**

To develop an AI/ML-enabled adaptive noise cancellation (ANC) system that effectively suppresses stationary, non-stationary, and impulsive defence noises while maintaining high speech intelligibility and real-time performance on embedded hardware

• **Theme- Miscellaneous**

• **PS Category- Hardware**

• **Team ID- SIH26-A0H-T333**

• **Team Name (Registered on portal) : Young Innovators**



MOTIVATION TO ADDRESS THE PROBLEM AND GENERATE SOLUTION

- To make communication clear even in noisy environments
- To reduce unwanted noise and make speech easier to hear
- To use AI/ML to handle different types of changing noise
- To provide real-time noise cancellation with very low delay
- To improve safety and communication for defence personnel

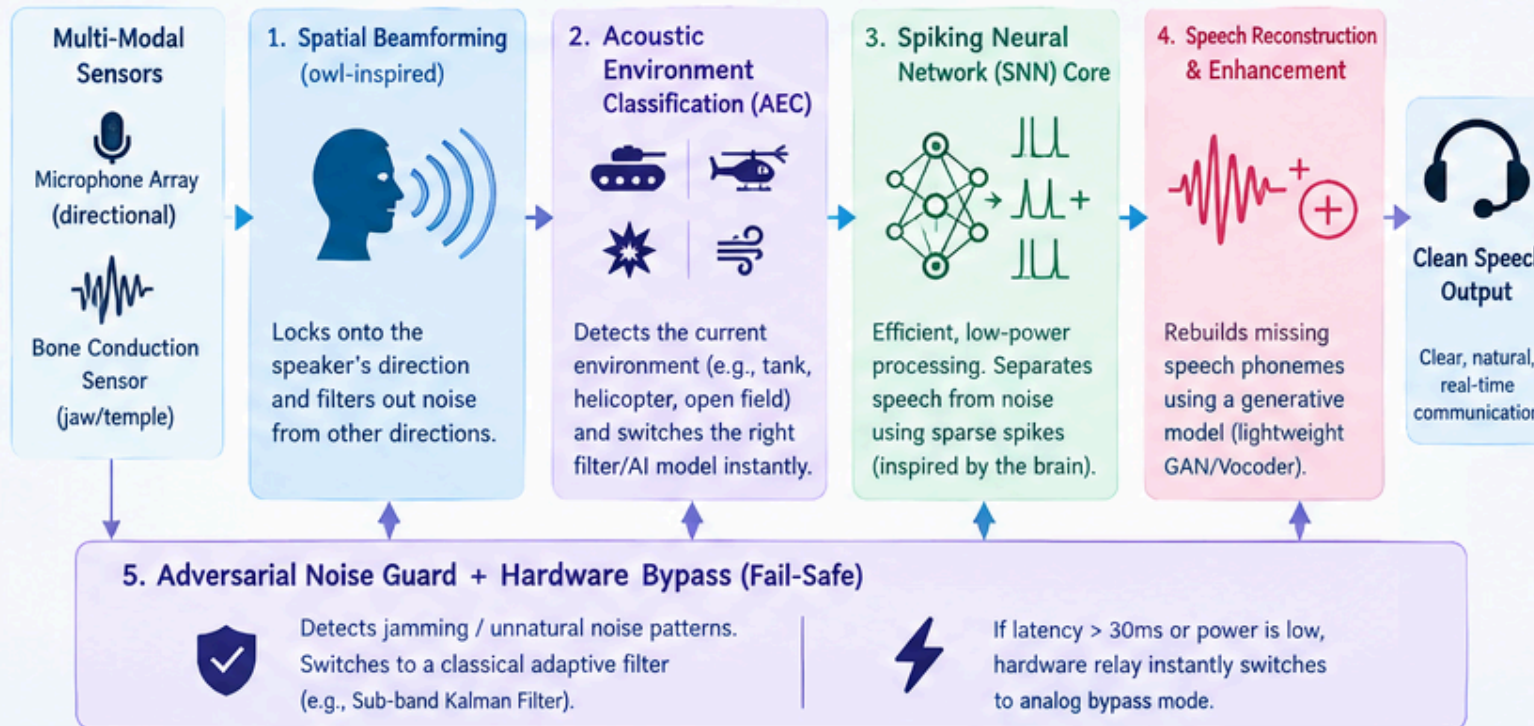
The Idea

BioHear+ uses nature-inspired principles and Spiking Neural Networks (SNNs) to understand the sound scene, separate the important voice, remove noise, and even reconstruct missing speech — all in real time, on a low-power embedded device.

Key Problems It Solves

- Loud and sudden noises (gunshots, explosions)
- Different noise types (vehicles, aircraft, wind, etc.)
- Voice loss during high noise
- Intentional jamming or interference
- System failure or high latency

How It Works



★ Why It's Unique & Feasible

- Combines biological principles with SNNs
- Handles multiple noise types and sudden impulses
- Reconstructs lost speech (not just cancels noise)
- Works on edge devices with very low latency (<5ms)
- Uses affordable, widely available hardware
- Can be built by a college team using open-source tools and libraries

The Tech Behind It

Spiking Neural Networks (SNNs)

- Event-driven (low power, low heat)
- Fast and efficient
- Handles real-time audio well

Traditional NN

Continuous computation



SNN

Event-driven (spikes only)



Implementation Stack

- 1 Train in Python**
Using SNN libraries (Singing/SnaX, Tonic, SNN Torch) with defence noise datasets.
- 2 Quantize & Export**
Convert to C++ using TFLite Micro or Edge Impulse.
- 3 Deploy on Hardware**
STM32H7 / Arduino Nano 33 BLE Sense / Raspberry Pi Pico (or similar).

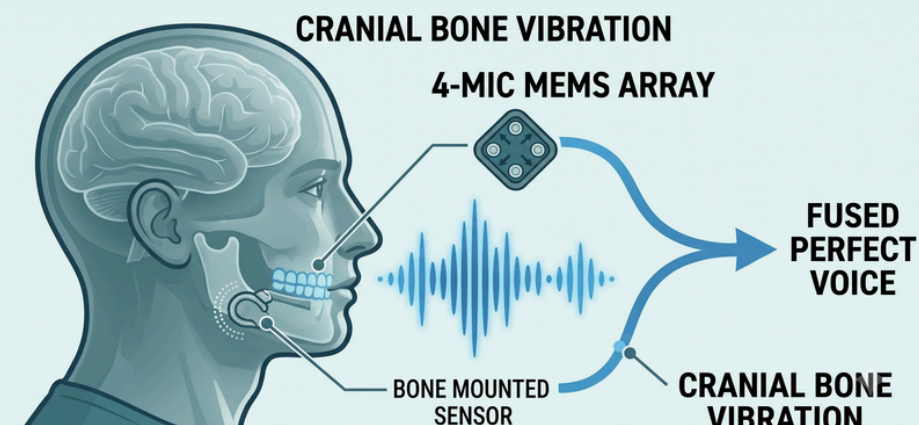
Prototype Hardware Setup



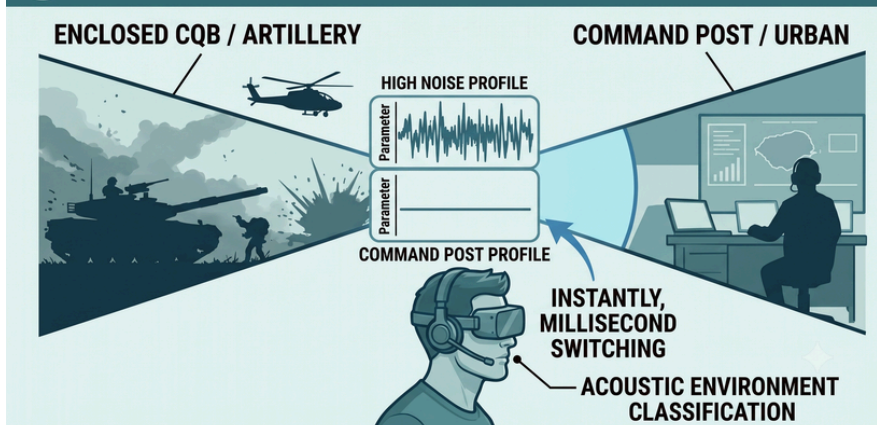
Expected Impact

- Clearer communication in critical missions
- Lower power & heat for longer use
- Real-time response (under 5ms)
- Higher accuracy in noisy and dynamic environments
- Safer and more efficient operations for defence personnel

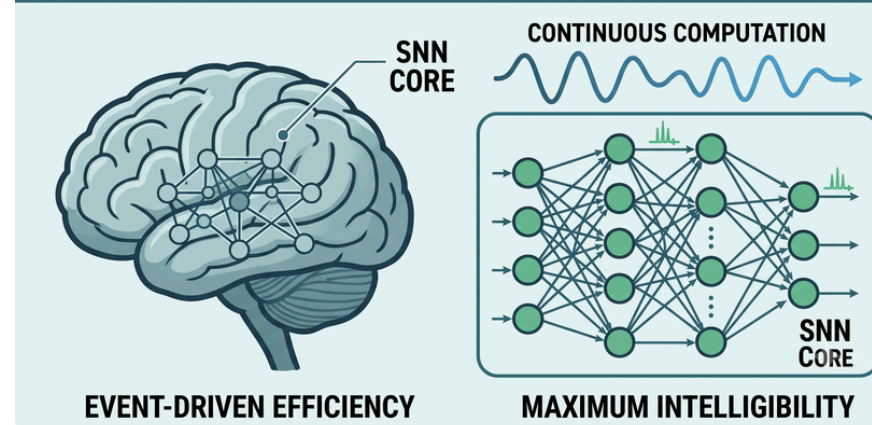
1 MULTI-MODAL HYBRID SENSING



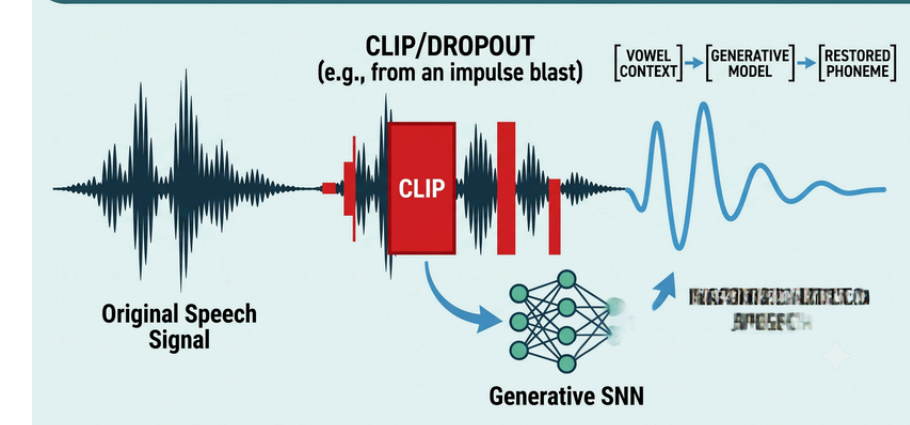
2 CONTEXT-AWARE DYNAMIC PROFILE SWITCHING (CADS)



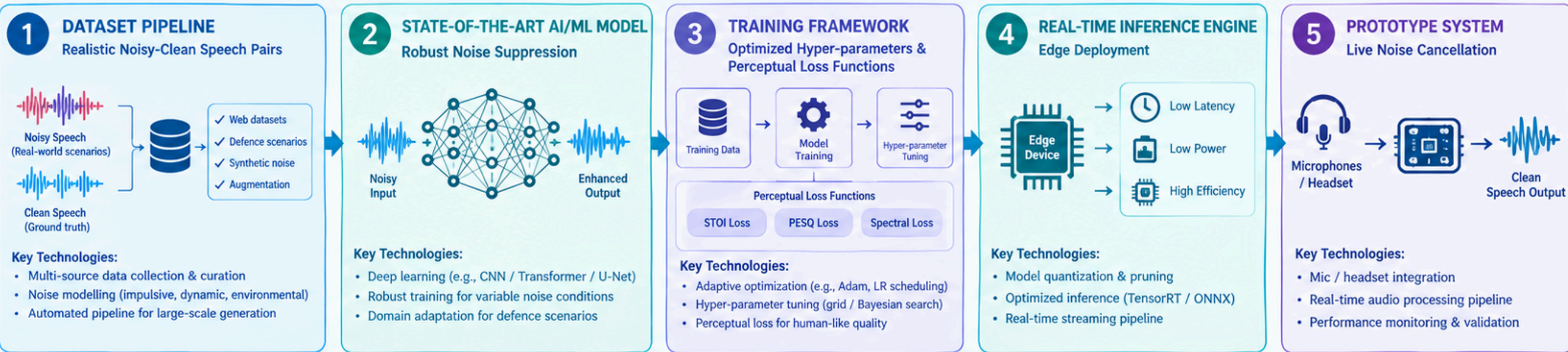
3 NEUROMORPHIC SNN CORE



4 GENERATIVE SPEECH RECONSTRUCTION



TECHNICAL APPROACH





TARGETED FUNCTIONALITIES

A hybrid AI-driven ANC system capable of real-time operation, handling impulsive and highly dynamic defence scenarios, and delivering high-quality, intelligible speech in extreme noise environments.



SCALABLE DATASET PIPELINE

Generates realistic noisy-clean speech pairs for diverse environments and scenarios.



STATE-OF-THE-ART AI/ML MODEL

Trained for robust noise suppression using advanced architectures and domain adaptation.



OPTIMIZED TRAINING FRAMEWORK

Uses hyper-parameter tuning and perceptual loss functions for better speech quality.



REAL-TIME INFERENCE ENGINE

Deployable on edge hardware with low latency and high efficiency.



PROTOTYPE SYSTEM

Live noise cancellation using microphones / headset integration.



TARGET PERFORMANCE

SNR > 15 dB	STOI > 0.85
PESQ > 2.5	Latency Low

 Enables reliable and intelligible communication in defence, aerospace and high-noise industrial environments.



FEASIBILITY AND VIABILITY



Commercial & Market Viability

- **Growing Demand:** Defence,Aerospace,police,paramilitary, industrial sectors.
- **Market size :** Global market for speech enhancement and ANC is expected to grow from approx \$2.5B (2024) to \$6.8B (2030).
- **Product Applications:** Headsets,vehicle intercom,aircraft,ground forces,disaster response.
- **Competitive Advantage :** AI+addaptive filtering,low cost ,high performance.

Technical Performance & Latency Metrics

- **Noise Suppression:** Handles stationary + impulsive noise (gunshots,explosions,vehicles)
- **Target Performance:** SNR>15dB, STOI>0.85, PESQ>2.5.
- **Low latency:**50 ms (real time communication).
- **Model:** CRN/DCCRN/Conv-TasNet/Transformer+LMS/NLMS(hybrid).
- **Features:** Time-frequency +complex domain,phase preservation,multi-band analysis.

Commercial & market oppurtunities

- Indian defence modernisation(Atmanirbhar bharat).
- Growing adoption of AI/edge devices in defence
- Potential for collabrations with DRDO,BEL,HAL.,etc.
- Scope for wearables,UAVs,and industrial communtcation systems.

Bill of Materials (BOM) & Cost Estimation for the Prototype Solution

Item /	Qty	Estimated	Total Cost
MEMS Microphone	1 set	₹1,200	₹1,200
Bone Conduction	1	₹800	₹800
Edge Processing	1	₹4,500	₹4,500
Audio Codec Module – High-	1	₹1,000	₹1,000
Headset Frame & 3D Printed	1	₹600	₹600
Recharge	1	₹900	₹900
Total			₹9,000

Strategic Risk & Insurance Feasibility

- Technical Risk: Model may underperform on unseen noise types.
- Hardware risk: Edge device limitations (quantization,pruning,TensorRT).
- Operation Risk : Microphone/communication integration and field conditions .
- Assurance: Redundancy(adaptive filter), fail-safe bypass, continuous monitoring.

OEM Integration & Operational Feasibility

- **Hardware :** NVIDIA Jetson Orin/AI SoC/DSP(edge deployment).
- **Software:**PyTorch,ONNX,TensorRT,Librosa,NumPy,MATLAB.
- **Integration:** Microphones (primary+references),hedset/communication unit,audio codec.
- **Feasibility:** High-standard hardware and open-source tools available .

National Security & Strategic Defence Benefits

- **Mission-Critical Reliability for Armed Forces:** Enables reliable, highly intelligible communication for military personnel operating in extreme defence, aerospace, and high-noise environments affected by heavy gunfire, artillery fire, and vehicle engines.
- **Indigenous Defence Innovation:** Replaces imported audio processing setups with local, domestic hardware and native neuromorphic software, building crucial indigenous intellectual property for India's defence sector.
- **Tactical Edge Optimization:** Leverages lightweight edge hardware and optimization techniques like quantization and TensorRT conversion to meet strict operational latency and power constraints in the field.

Socio-Economic & Public Safety Impact

- **Enhancing Public Safety Operations:** Directly supports high-noise emergency and security operations by filtering out chaotic acoustic disturbances like emergency sirens, heavy wind, and industrial chaos.
- **Low-Power and Sustainable Hardware:** Utilizes efficient, low-power neuromorphic SNN cores and widely available hardware components to ensure extended operational periods for frontline personnel.
- **Advanced Acoustic Benchmarks:** Delivers exceptional audio enhancement performance targeting SNR > 15 dB, STOI > 0.85, and PESQ > 2.5 to guarantee crystal-clear communications during critical missions.

Technological Integration & End-User Efficacy

- **Hybrid Multi-Modal Sensing:** Integrates a 4-mic MEMS array and cranial bone-conduction jaw sensors to fuse pristine input data even under severe dynamic interference.
- **Context-Aware Adaptability:** Utilizes Context-Aware Dynamic Profile Switching (CADS) to instantly adjust to changing environments—such as enclosed CQB, command posts, or urban sectors—in milliseconds.
- **Generative Speech Reconstruction:** Employs an on-device contextual AI vocoder to rebuild missing speech phonemes and lost information resulting from sudden noise bursts or signal drops.



RESEARCH AND RESOURCES



Government and Policy Research-

<https://idex.gov.in/idex>

<https://www.makeinindia.com/defense-acquisition-procedure>

<https://academic.oup.com/milmed/article/183/9-10/e547/4841664>

<https://www.audiology.org/news-and-publications/audiology-today/articles/military-service-and-hearing-health-the-noise-study/>

<https://nap.nationalacademies.org/catalog/11443/noise-and-military-service-implications-for-hearing-loss-and-tinnitus>

Commercial Software and HIL Hardware Tooling

<https://www.speedgoat.com/solutions/industries/consumer-electronics/audio-simulation-control-and-testing>

<https://www.mathworks.com/hardware-support/real-time-simulation-and-testing-with-speedgoat-hardware.html>

<https://www.dspace.com/en/inc/home/products/hw/scalexio.html> https://www.mathworks.com/products/connections/product_detail/dspace-hil-test.html <https://www.ap.com/analyzers-accessories/apx500/> <https://www.ni.com/en/solutions/aerospace-defense/embedded-software-test.html>

Biomimetic and Robotics Research Publications

<https://iopscience.iop.org/journal/1748-3190> <https://www.science.org/journal/scirobotics> <https://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=8613444>

<https://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=8860> <https://www.frontiersin.org/journals/bioengineering-and-biotechnology/sections/bionics-and-biomimetics> <https://www.nature.com/natmachintell/>

SOME MISCELLANEOUS

<https://github.com/jeshraghian/snntorch>

<https://github.com/ARM-software/CMSIS-DSP> , <https://zenodo.org/records/1227121>

<https://github.com/microsoft/DNS-Challenge>, <https://github.com/aliutkus/speechmetrics>

<https://www.itu.int/rec/T-REC-P.862>, https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=35978